

Radiological Characterization of Malformations of the Internal Auditory Canal

Radiologische Charakterisierung von Fehlbildungen des inneren Gehörgangs

Authors

Katja Döring¹, Rolf Salcher², Heinrich Lanfermann¹, Thomas Lenarz², Athanasia Warnecke², Anja Gieseemann¹

Affiliations

- 1 Department of Diagnostic and Interventional Neuroradiology, Hannover Medical School, Hanover, Germany
- 2 Department of Otorhinolaryngology, Hannover Medical School, Hanover, Germany

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Correspondence

Katja Döring

Department of Diagnostic and Interventional Neuroradiology, Hannover Medical School, Hanover, Germany
doering.katja@mh-hannover.de

ABSTRACT

Purpose Malformations of the internal auditory canal (IAC) are rare. They can present as a narrowing of the canal due to aplasia or hypoplasia of the vestibulocochlear nerve, complete atresia, or even a doubling of the IAC. The aim of our study is to provide a comprehensive overview of the different types of IAC anomalies and to provide a classification based on radiological findings and their relation to syndromes and/or inner/middle ear anomalies. In addition, IAC malformations will be explained in a putative phylogenetic context.

Material and Methods All patients who underwent pre-interventional imaging between 1995 to 2024 for evaluation of a hearing implant in situ with an inner ear malformation were included in the present retrospective study. The available imaging data, i. e. high resolution computed tomography (HRCT) or cone beam CT (CBCT) of the temporal bone and

supplementary MR of the temporal bone, were reviewed independently by two neuroradiologists. Malformations of the IAC and concomitant malformations of the middle and inner ear were recorded. Demographic and clinical data were also collected. Based on the data and information obtained, a radiological classification of the different IAC malformations was provided.

Results A total of 36 patients (55 affected ears) were included in the analysis. The majority of the patients were female (75%). The mean age was 6.3 ± 9.4 years (mean $\pm$ std). A syndromic disease was present in 28% of the patients. Due to severe hearing loss, a total of 48% of patients received a hearing system. Based on the radiological findings, we performed the following typing: Type I – Narrow IAC; Type II – Atresia with isolated facial nerve canal; Type III – Double IAC with/without atresia, Type IV – Complete atresia. In descending order, the frequency of these malformations of the IAC in our cohort was distributed as follows: Type III – Double (n=29, 52.7%), Type I – Narrow (n=15, 27.3%), Type IV – Complete atresia (n=4, 7.3%), Type II – Atresia with isolated facial nerve canal (n=7, 12.7%).

Conclusion Since hypoplastic IAC may be associated with hypoplastic or absent cochlear nerve and sensorineural hearing loss, radiological assessment of the IAC is of critical importance in the evaluation of patients with severe sensorineural hearing loss undergoing cochlear implantation. Accurate analysis of the imaging data and understanding the complexity of the malformations are of great importance in assessing the expected benefits prior to cochlear implantation.

Key Points

- IAC malformations can be classified into four different groups based on recurring patterns.
- IAC malformations are often associated with hypo-/aplasia of the vestibulocochlear nerve.
- The facial nerve is usually present, but may have an aberrant course.

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ZUSAMMENFASSUNG

Ziel Fehlbildungen des inneren Gehörgangs (IAC) sind selten: Sie können sich als Verengung des Gehörgangs aufgrund einer Aplasie, als vollständige Atresie oder sogar als Verdopplung des IAC präsentieren. Ziel dieser Studie ist es, einen umfassenden Überblick über die verschiedenen Arten von IAC-Malformationen zu geben und eine Klassifizierung auf Basis radiologischer Befunde und ihrer Assoziation mit Syndromen und/oder Innen- bzw. Mittelohrfehlbildungen zu erstellen. Darüber hinaus werden IAC-Fehlbildungen in einem phylogenetischen Kontext erläutert.

Materialien und Methoden In die vorliegende retrospektive Studie wurden alle Patienten aufgenommen, die zwischen 1995 und 2024 zur Beurteilung eines Hörimplantats bei einer Innenohrfehlbildung einer präinterventionellen Bildgebung durchlaufen haben. Die verfügbaren Bildgebungsdaten, hochauflösende Computertomografie (HRCT) oder digitale Volumentomografie (DVT) des Felsenbeins sowie ergänzende MRT-Aufnahmen des Felsenbeins, wurden von zwei Neuroradiologen unabhängig voneinander ausgewertet. Dabei wurden Fehlbildungen der IAC und begleitende Fehlbildungen des Mittel- und Innenohrs dokumentiert. Zudem wurden demografische und klinische Daten erhoben. Auf der Grundlage der gewonnenen Daten und Informationen wurde eine

radiologische Klassifizierung der verschiedenen IAC-Fehlbildungen vorgenommen.

Ergebnisse Insgesamt wurden 36 Patienten (55 betroffene Ohren) in die Analyse einbezogen. Die Mehrheit der Patienten war weiblich (75%). Das Durchschnittsalter betrug $6,3 \pm 9,4$ Jahre (Mittelwert $\pm$ Standardabweichung). Bei 28% der Patienten lag eine syndromale Erkrankung vor. Aufgrund eines schweren Hörverlusts erhielten 48% der Patienten ein Hörsystem. Auf der Grundlage der radiologischen Befunde erfolgte folgende Typisierung: Typ I: enge IAC; Typ II: Atresie mit isoliertem Facialiskanal; Typ III: doppelter IAC mit/ohne Atresie; Typ IV: vollständige Atresie. In absteigender Reihenfolge verteilte sich die Häufigkeit dieser Fehlbildungen der IAC in unserer Kohorte wie folgt: Typ III (doppelter innerer Gehörgang): $n=29$ (52,7%), Typ I (enger innerer Gehörgang): $n=15$ (27,3%), Typ IV (vollständige Atresie): $n=4$ (7,3%), Typ II (Atresie mit isoliertem Fazialiskanal): $n=7$ (12,7%).

Kernaussagen

- IAC-Fehlbildungen lassen sich anhand wiederkehrender Muster in vier verschiedene Gruppen einteilen.
- IAC-Fehlbildungen gehen häufig mit einer Hypo-/Aplasie des Nervus vestibulocochlearis einher.
- Der N. facialis ist in der Regel vorhanden, kann aber einen aberranten Verlauf aufweisen.

Abbreviations

ABI	auditory brain implant
BOR	branchio-oto-renal syndrome
CI	cochlea implant
HRCT	high resolution computed tomography
CBCT	cone beam computed tomography
cpa	cerebellopontine angle
FN	facial nerve
IAC	internal auditory canal
IPT 1	cochlear incomplete partition type 1
MRI	magnetic resonance imaging
NCC	neural crest cells
HRCT	high-resolution computed tomography
SCC	semicircular canal

Introduction

Malformations of the internal auditory canal (IAC) are rare and receive less attention than malformations of the inner ear. IAC malformations can occur in isolation or in conjunction with malformations of the inner ear. The most severe inner ear malformations, such as cochlear aplasia or complete aplasia of the semicircular canals, which are often accompanied by nerve hypoplasia or aplasia, show a narrowed or otherwise altered appearance of the IAC. The present article focuses on isolated cases of IAC malformations in which the cochlea, vestibule, and semicircular canals (SCC) ap-

pear largely normal on a temporal bone CT. The majority of extant literature on the subject consists of case reports [1, 2, 3, 4, 5, 6, 7, 8] that describe narrow, atretic, and double internal auditory canals. A narrow canal is typically characterized as an IAC with a central width of 2 mm or less, manifesting either in isolation or in conjunction with a malformation of the inner ear or syndromic hearing loss. Double IACs may open into the cerebellopontine angle (cpa), or one canal may open into the cpa with the other opening closing towards the cpa and tapering into a conical or atretic shape.

The majority of cases documented in the extant literature concerning IAC malformations report normal facial nerve function. Other case reports have documented cases of facial synkinesia and complete facial nerve palsy. The course of the facial nerve is known to be highly variable, especially in the middle ear, which complicates surgery in this area [9]. Aplasia of the auditory nerve with a concomitant malformation of the internal auditory canal provides information about the development of the cranial nerves and probably also about further changes in the formation of the brain stem/brain nuclei themselves.

The objective of the present study is therefore to record all IAC anomalies, categorize and classify them according to recurring patterns, document the respective course of the facial nerve canal and discuss the results in a phylogenetic context. A preoperative understanding of the accompanying anomalies of the temporal bone, such as malformations of the outer, middle, and inner ear and the migration of the facial nerve (FN) into the auditory canal, is of great benefit to otologists in developing appropriate surgical approaches for patients with IAC malformations who are candi-

dates for cochlear implants [8]. The present study constitutes the most extensive collection of its kind to date, facilitating the categorization of imaging features associated with IAC malformations and related syndromic disorders (e.g. branchio-oto-renal (BOR) syndrome [2]). This study offers valuable insights for otologists and it aims to raise awareness among (neuro)radiologists regarding the intricate anatomical space between the CPA and the inner ear, which is characterized by its complexity and small size.

Material and Methods

Our ethics committee approved this retrospective study and waived the requirement for informed patient consent.

Patient selection

In a tertiary referral center, all patients were referred from the Department of Otolaryngology for assessment of sensorineural hearing loss and evaluation for cochlear implantation. This usually involved imaging, including a cone beam or high resolution computed tomography (CB-CT or HR-CT) of the temporal bone and an MRI of the temporal bone. All patients treated between 1995 and 2024 were screened and only those who showed an isolated IAC malformation on imaging were included in the study, regardless of age or gender. In addition to the presence of cochlear aplasia, aplasia of the SCC was also an exclusion criterion. Demographic data and additional clinical information, including the presence of facial nerve palsy, syndromic disease, hearing aid fitting, etc., were added.

Imaging of the temporal bone

All patients underwent HRCT or CBCT of the temporal bone. Different types of scanners (HiSpeed Advantage RP, HiSpeed Advantage and LightSpeed16; all GE, Milwaukee, WI, USA) were used over the long period of our study and slice thickness ranged from 0.625 to 1 mm. The acquired images were uploaded to our current PAC system (GE, Milwaukee, WI, USA), with the exception of four patients for whom only the plain films were available. In the majority of cases (n = 29), complementary MRI of the head and temporal bone was available, acquired with three different 1.5 T MR systems (Signa Horizon, GE, Milwaukee, WI, USA (16 cases); Avanto and Aera, Siemens medical systems, Erlangen Germany (8 cases)) and a 3 T System (Verio, Siemens medical systems, Erlangen Germany, 10 cases). Scan protocols included axial T2-weighted FLAIR (TR 9000; TE 172; TI 2200; FOV 24 × 24; slice thickness 6 mm; distance 1.5 mm; matrix 256 × 192), sagittal T2-weighted fast spin echo (TR 6060; TE 95; FOV 24 × 24; slice thickness 3 mm; distance 0.3 mm; matrix 512 × 256), axial T1-weighted spin echo (TR 500; TE 14; FOV 24 × 18; slice thickness 3 mm; distance 0.3 mm; matrix 320 × 256) and a T2-weighted three-dimensional gradient echo sequence (TR 11; TE 5; FOV 16 × 16; slice thickness 0.8 mm; flip angle 65°; matrix 512 × 512) of the cerebellopontine angle region. Reconstructions perpendicular to the internal auditory canal (IAC) were made in all but two cases.

Study design and radiological characterization of malformations of the internal auditory canal

The diagnosis of IAC malformations and the assessment of the nerves were made by two neuroradiologists by consensus. The various malformations of the internal auditory canal were classified based on recurring patterns as follows:

- **Type I: Narrow.** Normal localization of the IAC with a width ≤ 2 mm.
- **Type II: Atresia of IAC with isolated facial nerve canal.** Always accompanied by complete atresia of the IAC. Anteriorly displaced canal, directly continuous with the pars labyrinthi/bony part of the facial nerve canal (running above the cochlea). MRI shows one nerve in the cpa, running into the canal. A connection to the fundus area may or may not be present.
- **Type III: Double IAC with/without atresia.** Two canals can be identified. The antero-superior canal was always open into the cerebellopontine angle (cpa). The postero-inferior canal showed atresia in more than the half of cases. The postero-inferior canal may contain rudimentary nerve fibers.
- **Type IV: Complete Atresia.** No canal or only a discontinuous canal can be identified, often tapering towards the cpa. If a facial canal is present see Type II.

In addition, the communication of the canals at the fundus of the internal auditory canal was listed. The presence of the vestibulocochlear nerve was recorded as seen on MRI, if available. The course of the facial nerve was assessed in all relevant segments: the intracranial, labyrinthine, tympanic, and mastoid parts were examined for presence, deviation, and duplication. Clinical history was obtained from the patient's medical records. Information on facial and vestibulocochlear nerve function was collected. Implantation of a cochlear implant (CI) and auditory brainstem implant (ABI) was documented, and the outcome was noted.

Means of assessing intracranial structures

The intracranial portion of the facial nerve was identified as such if it could be traced from the brainstem to the entrance of the bone and a small canal towards the facial ganglion could be recognized. The presence of the facial nerve in the middle ear was accepted if the tympanic portion could be identified. Atresia of the IAC was considered round if no canal or part of a canal was found towards the intracranial border of the temporal bone. The vestibulocochlear nerve was labelled absent/aplastic if no continuous nerve structures were seen within the auditory canal/the cpa. Nomenclature: The term canal of facial nerve is used for the bony canal through the temporal bone containing the facial nerve (in the case of duplication or atresia). In duplication, it is always the antero-superior canal of the two canals. In the case of atresia, it mostly has a more anterior course, it may even start at the trigeminal entry into the bone and then leading to the gasserian ganglion, coinciding with the well-known curved labyrinthine segment of the facial nerve. In cases of duplication (in the literature and in our case series) there is always an antero-superior canal, usually small but always open, containing the facial nerve. In some duplicated cases two nerves may enter the anterior canal. The other canal is postero-inferior and may be open, narrow, or stenotic. In none of

the cases in our series or in the literature was a continuous vestibulocochlear nerve identified within this canal. However, non-continuous nerve structures can be identified in some atretic shy;canals.

Results

Baseline characteristics of the study population

A total of 36 patients (59 ears) with isolated IAC malformation were included in the analysis. The age of the patients ranged from 5 months to 75 years. On average, the collective was very young with an age of 6.3 ± 9.4 years (mean $\pm$ std). Nine patients were male and 27 are female. In 19 cases both ears were affected, in 17 cases the IAC malformation is unilateral. An underlying syndromic disease was identified in 28 % (10/36) of the patients, including Goldenhar syndrome (1/10), trisomy 21 (1/10), Kippel-Feil syndrome (1/10), Nagel syndrome (1/10), but also muscle hypotonia syndromes (2/10) and a further four suspected cases of syndromic diseases whose genetic clarification is pending. 38.8 % received a hearing system, with cochlear implants (CI) being by far the most common type of implant (27.7 %, for further details see ► Table 1).

Distribution of IAC malformations

Double IACs were the most frequently diagnosed malformations in our group ($n = 29$, 52.7 %), with double IACs with atretic course of a single canal being significantly more common. The other types of IAC malformations were distributed as follows in descending order of frequency (► Table 2 , ► Table 3 , ► Fig. 1, ► Fig. 2, ► Fig. 3, ► Fig. 4): Narrow IAC ($n = 15$, 27.3 %) – Complete Atresia ($n = 4$, 7.3 %) – Atresia of IAC with isolated facial nerve canal ($n = 7$, 12.7 %). Three patients in our group had facial nerve palsy. In only one case was the facial nerve not visible on imaging. Within the different IAC malformations, nerve structures have been imaged with varying frequency. ► Table 4 and ► Table 5 provide detailed breakdowns.

Hearing performance II (CAP-II) after implantation of a hearing system

Follow-up responses after implantation were obtained for three of the four patients who received an ABI. Two improved to a CAP-II score of 2 or 3, while one improved to a score of 1 (see also ► Table 5 and ► Table 6). Patients with a CI showed a very positive response to the implant during follow-up. Of the ten patients with a CI, two increased their score to 9, two to 8, two to 6, and one to 7 (► Table 7).

► Table 1 Baseline data.

Label	Numbers	Percentage of total combined	Description
patient cohort	36		
sex	9/27	25%/75 %	(m/w)
age	6.3±9.4		mean ± std
total ears affected	59		
bilateral	23	60 %	
unilateral	13	40 %	
Syndromic disease, a.o.	10	28 %	
Goldenhar syndrome	1		
Trisomy 21	1		
Kippel-Feil syndrome	1		
Muscle hypotonia syndromes	2		
Nager syndrome	1		
other not specified	4		
Hearing System		38.8 %	
Cochlear Implant (CI)	10	27.7 %	
CI unilateral	8		
CI bilateral	2		
Central auditory implants	4	11.1 %	
ABI unilateral	4		

Discussion

Isolated IAC malformations are rare, and available publications are typically limited to case reports [10, 11, 12, 13]. Common classifications of IAC malformations describe these as part of complex inner ear malformations, including cochlear atresia, IPT I, or complete atresia of the semicircular canals [10, 11]. To our knowledge,

this is the largest series on isolated IAC malformations reported thus far. Based on the analysis of 55 affected ears, we were able to categorize four types of isolated IAC malformations: complete atresia, doubled IAC, narrow IAC and atresia of IAC with isolated facial nerve canal. The doubled internal auditory canal was further subdivided according to an open or atretic posterior canal. Further neural structures in the atretic canal and a connection of the canals at the fundus were noted.

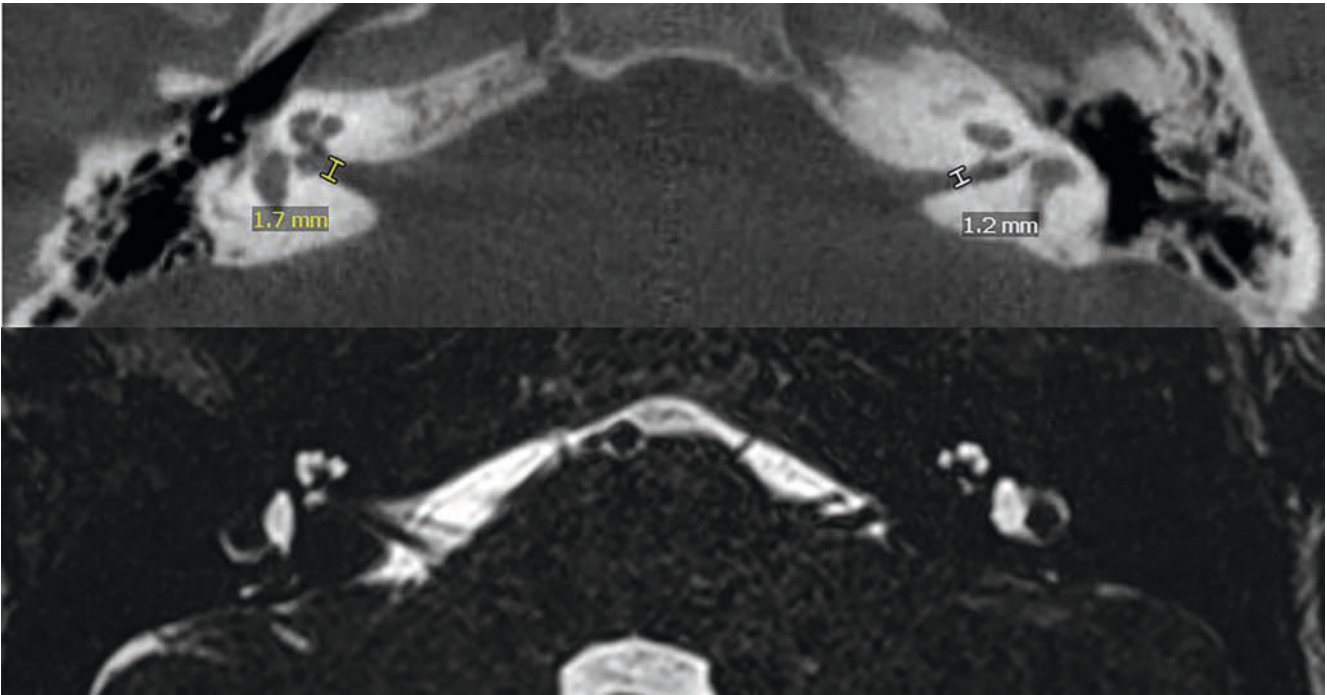
One of the few studies that corroborates our findings is a series describing isolated IAC malformations that was published by Wang et al. in 2019 [2]. Similar to our observations, the authors divided isolated IAC malformations into duplications of the canal isolated by complete or incomplete bony septa, whose orientation varies from horizontal to almost vertical. The facial nerve (FN)

► **Table 2** Type of IAC malformation.

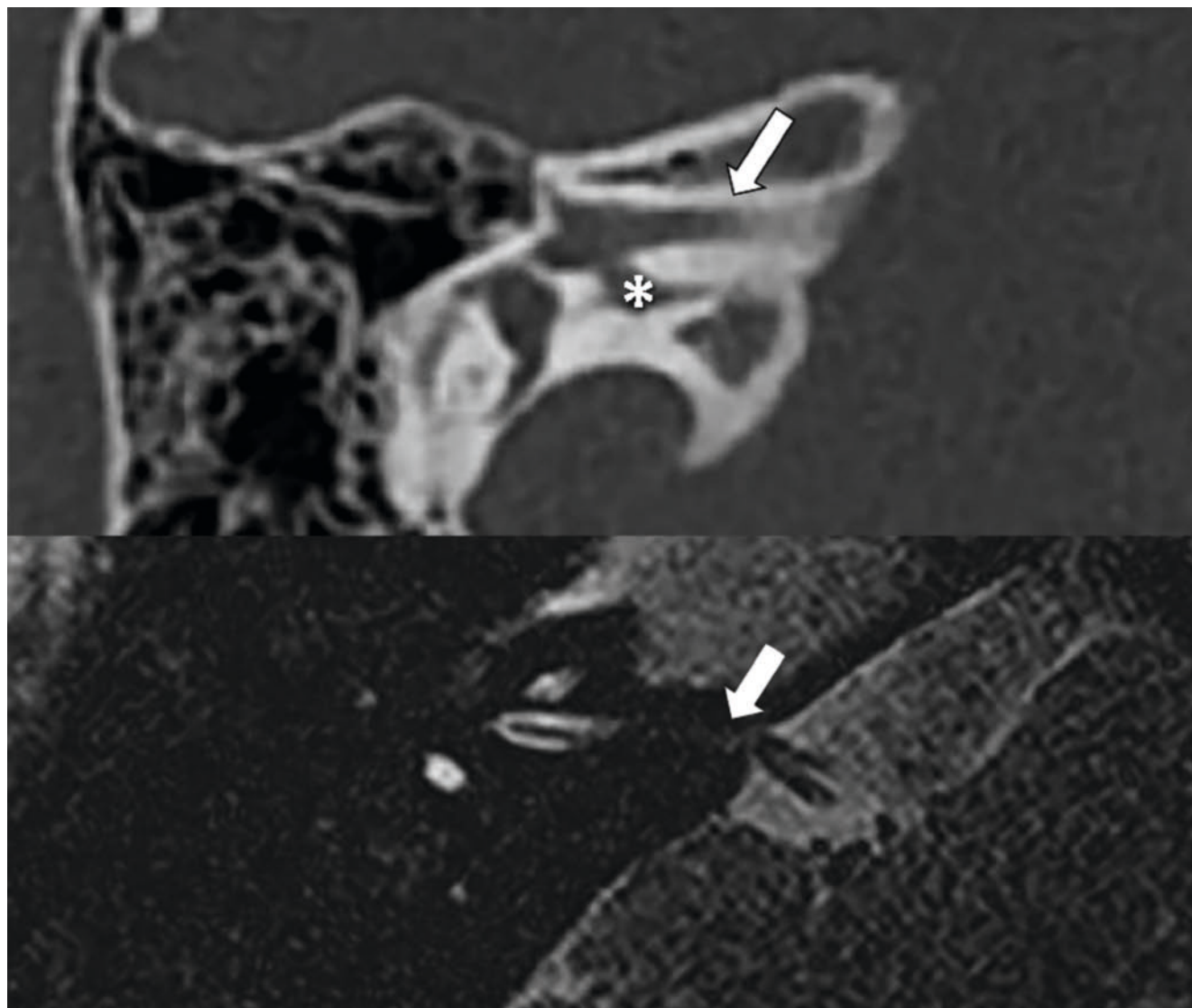
	Type	Definition/criteria
I	Narrow IAC	Normal location, greatest width ≤ 2 mm
II	Atresia of IAC with isolated facial nerve canal	Anterior displaced location and directly continuous with the pars labyrinthi/bony part of the facial nerve canal (running above the canal)
III	Double	
III	▪ with atresia of the IAC (IIIa)	Two canals of which the dorsal one is only partially existent, tapering towards the CPA
	▪ without atresia of the IAC (IIIb)	Two canals that start together at the fundus and open in two parts towards the CPA
IV	Complete Atresia	Canal is not or only partially existent, tapering towards CPA

► **Table 3** Distribution of IAC malformation.

	Type	Count of affected ears
I	Narrow	15
II	Single canal of the facial nerve	7
III	Double	29
	▪ with atresia of the IAC (IIIa)	20
	▪ without atresia of the IAC (IIIb)	9
IV	Atresia	4



► **Fig. 1** Upper image – HRCT of the temporal bone, transverse plane. IAC type I: The IAC is primarily narrowly configured on both sides. On the right, the IAC measures a maximum of 1.7 mm; on the left, it is even narrower at 1.2 mm. Lower image: MRI, T2 SPACE in transverse plane. Same patient. Due to the narrow location of the IAC on both sides, the vestibulocochlear and facial nerves can only be clearly distinguished in the cpa; further intrameatal differentiation is not possible despite the use of a high-resolution MRI.



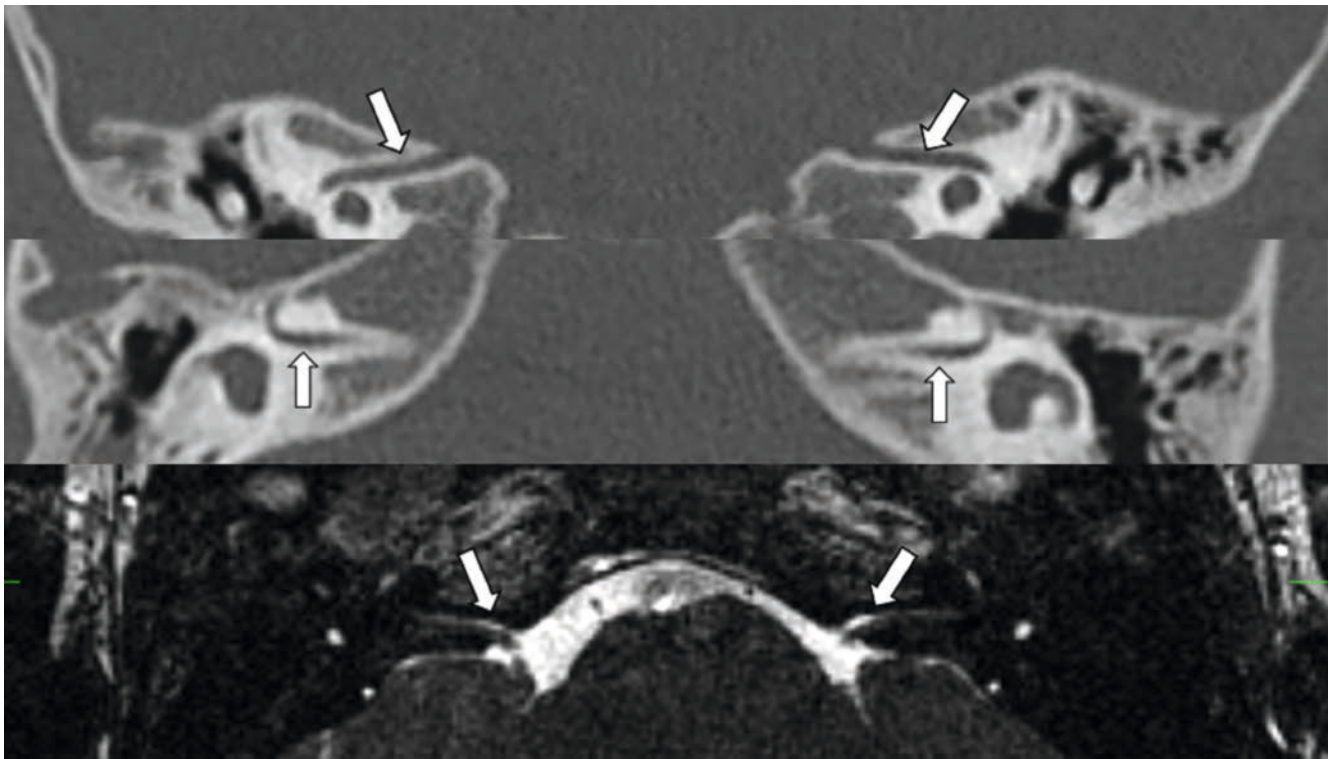
► **Fig. 2** Upper image – HRCT of the temporal bone. Section of the right temporal bone in coronal plane. IAC type III – double IAC with atretic postero-inferior canal that has no opening to the CPA. The white arrow marks the anterosuperior canal, the star marks the postero-inferior canal. Lower image – MRI, T2 SPACE in transverse plane. The arrow marks the nerve structures running in the antero-superior canal.

mostly passes through the anterior superior canals, similar to our observations. The postero-inferior canals are partially blind or were open as the anterior superior canals to the CPA. Demir et al. [12] and Baik et al. [13] also discussed duplicated canals, in the form of case reports. As the IAC in each case consisted of two separate narrow bone canals, both authors referred to a narrowed duplicated IAC. Demir et al. reported a 7-year-old boy with associated Klippel-Feil syndrome and a narrow double IAC without sensorineural hearing loss but with conductive hearing loss [10]. In this patient, the IAC consisted of two separate narrow canals that were clearly visible on a 3D CT scan of the temporal bone and a nerve that was visible on MRI.

Baik et al. presented the case of a 6-year-old girl with unilateral hearing loss [13]. CT also showed a narrow double canal of the IAC. The IAC was divided by a bony septum into a relatively large anterior upper part and a narrow posterior lower part. The ante-

rior upper part ended in a wide connection to the facial nerve canal and a narrow connection to the vestibule. The posterior lower part ended in narrow connections to the cochlea and the vestibule. The two canals partially joined at the lateral end of the right IAC. The facial nerve canal was intact along its course except for a slightly dilated labyrinthine portion. Supplementary MRI showed no identifiable nerve structure in the infero-posterior compartment of the right double IAC. There was only one nerve structure in the anterosuperior compartment of the double IAC, presumably the facial nerve.

The numerous but recurrent patterns that allow for classification of IAC malformations raise the question of which phylogenetic processes are responsible for nerve branching and at what stage of embryogenesis an interruption occurs. There are two widely accepted hypotheses that attempt to explain the relationship between IAC malformation and sensorineural hearing loss.



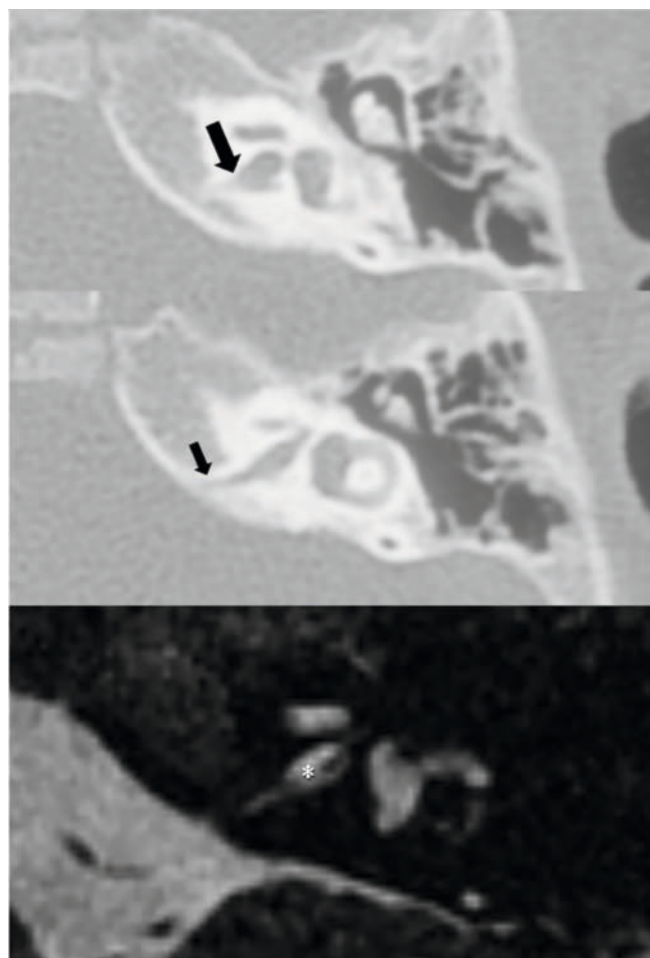
► **Fig. 3** The top two images are HRCT scans of the temporal bone. The top image is in the coronal plane, while the middle image is in the transverse plane. A Type II IAC malformation is present on both sides, with complete atresia of the IAC and an isolated facial nerve canal. The upper image clearly shows the lack of connection between the facial canal and the cochlea. The bottom image shows the situation in an MRI T2 SPACE scan in the transverse plane and illustrates the facial nerve emerging from the CPA. The white arrows in the two top images indicate the isolated course of the facial nerve canal. The facial nerve is marked with white arrows in the bottom image.

It is thought that unlike the inner ear, which develops from the otocyst (a thickening and invagination of the otic placode) [14], the bony IAC develops from chondrification of the mesoderm surrounding the facial and vestibulocochlear nerves. These nerves originate from the facial-acoustic primordium, a neural crest derivative, and can be identified as two separate nerves around the sixth week of gestation. The mesenchyme surrounding these nerves undergoes chondrification and later ossification to form the IAC during the fifth and sixth months of gestation [15]. The first hypothesis is based on the assumption that the embryonic cochlea and vestibule induce the growth of the vestibulocochlear nerve and that the bony canal develops with the facial nerve through chondrification and ossification of the mesoderm surrounding the nerve. The hypothesis assumes that if the growth of the vestibulocochlear nerve ceases, the development of the surrounding IAC will also stop or is getting narrow if the vestibulocochlear nerve is hypoplastic [15, 16]. The other hypothesis is that a bony stenosis of the IAC is the cause, inhibiting the growth of the vestibulocochlear nerve and causing impaired transmission of the induction signal from the intact cochlea and vestibule. However, the preservation of facial nerve function and normal macroscopic morphology in most cases of narrow IAC makes this hypothesis less likely [15].

A new approach to explain the developmental abnormalities of the IAC is based on the assumption that the interactions between glial cells and neural progenitor cells may be disrupted during the

development of the vestibulocochlear nerve. To test this hypothesis, Sandell et al. [17] examined the morphological development of the mouse vestibulocochlear nerve in relation to the two embryonic cell types from which it arises: the otic vesicle-derived precursor cells, which give rise to neurons, and the neural crest cells (NCC), which give rise to glial cells. They found that disruption of NCCs derived from rhombomere 4 (R4) impaired the formation of the central axonal projections of the vestibulocochlear nerve [17]. Schlosser et al. made similar observations, but focused on disruption of ErbB2, a receptor that mediates the interaction between neurons and glial cells via the ligand neuregulin 1 [18]. Mice with ErbB2 mutations showed impaired migration and survival rates of Schwann cell precursors. In the auditory nerve, loss of ErbB2 leads to abnormal migration of spiral ganglion neurons, abnormal alignment of peripheral neurites and a reduced number of neurons [19, 20]. These findings support a model in which the CV nerve, like other sensory cranial nerves, develops through integrated interactions between neuronal progenitors derived from placodes and glial progenitors derived from NCCs. Disruption of any of these components leads to defective/impaired nerve development.

Despite the expectation of a high incidence of facial nerve palsies in patients with complete IAC atresia, to the best of our knowledge, only three cases of facial nerve palsy in addition to an IAC malformation have been reported thus far [2, 3, 4]. This finding aligns with the results of the present study, wherein only



► **Fig. 4** The two upper images show an isolated section of the left temporal bone in the transverse plane. The images are taken at different heights and show the fundus of the IAC (upper image) and the atretic end (middle image), with no opening to the CPA (marked by black arrows). The bottom image shows the situation in an MRI T2 SPACE scan. Rudimentary parts of the vestibulocochlear nerve can be seen in the fundus (marked by the white star). No nerve structures are visible in the cpa.

three patients (8.3%) exhibited facial nerve palsy. However, it is important to note that a clear allocation to a specific type of IAC malformation is not possible, as the prevalence of facial nerve palsy is distributed across all types of IAC malformations. In an attempt to reconcile the heterogeneous picture of different IAC malformations with the equally mixed picture of different facial canal trajectories, the perspective is limited to the etiologically unexplained Möbius sequence (congenital oculofacial paralysis). This rare congenital disorder, characterized by paralysis of the sixth and seventh cranial nerves, occurs in 2–20 cases per 100,000 births and may be observed within affected families [20, 21, 22, 23]. Genetic factors and teratogenic substances have been suggested as possible causative agents. The potential pathological role of ischemia in the brainstem nuclei is also a subject of discussion. This finding suggests the hypothesis that axon growth from the lateral to the brainstem determines the osseous development of the IAC, although this remains to be proven through embryological evidence. In the event of the nerve fibers terminating in the tapered canal, it can be deduced that the signals from the brainstem have been interrupted. This is presumably due to the destruction of the cells of the cochlear/vestibular nerves to varying degrees. In the case of complete atresia, i.e. premature interruption in the embryonic phase, the axons are unable to move in the direction of the brainstem. Consequently, in the event of the presence of the facial nerve and its nucleus, the nerve has the capacity to establish its own channel. The purpose of these hypotheses is to explain the diverse and heterogeneous picture of IAC malformations. Further research is required to substantiate this claim. This should include the development of robust and validated animal models. It can be hypothesized that the disruption of the development of the vestibulocochlear nerve (i.e. denervation) leads to secondary changes in bony structures, depending on the time of interruption of embryogenesis. In cases where the displacement of the facial nerve is rostrally, in conjunction with the presence of a separate course and an empty atretic inferior alveolar canal (IAC), the interruption of embryogenesis at an early stage may be the underlying cause. Notwithstanding the interruption to embryogenesis, the facial nerve will only form its own canal if the nerve and its nucleus are present.

► **Table 4** Numbers and percentage of visible nerves in the cerebello pontine angle (CPA) and IAC listed by the type of inner ear malformation.

Type of IAC malformation		Number of ears	Visible nerves in cpa			Visible nerves in IAC				
			0	1	2	0	1	2	3	4
I	Narrow	15	33%	7%	60%	27%	7%	7%		13%
II	Single canal of facial nerve	7	17%	83%		100%				
III	Double	29 (with atresia 20, without atresia 9)	3%	47%	40%	47%	3%	22%	9%	6%
IV	Atresia	4	83%	17%		50%	50%			

► **Table 5** Categories of Auditory Performance II (CAP-II) Score.

CAP-II Score	Corresponding Skill
0	No awareness of environmental sounds or voice
1	Awareness of environmental sounds
2	Response to speech sounds
3	Identification of environmental sounds
4	Discrimination of speech sounds without lip reading
5	Understanding of common phrases without lip reading
6	Understanding of conversation without lip reading
7	Use of telephone with known speaker
8	Follows group conversation in a reverberant room or where there is some interfering noise, such as classroom or restaurant
9	Use of telephone with an unknown speaker in an unpredictable context

► **Table 6** CAP-II-Outcome of patients who received an ABI.

Pt.no.	IAC malformation	Laterality of ABI	Description	Post-OP CAP-II*
1	Type IV right, Type IIIa left	Unilateral/left ear		1
2	Type IIIa both sides	Unilateral/left ear		3
3	Type IIIa both sides	Unilateral/left ear	Not available for follow up	n/a
4	Type II right, Type IIIa left	Unilateral/left ear		2

*Categories of Auditory Performance II (CAP-II) Score. ABI – Auditory Brainstem Implant.

► **Table 7** CAP-II-Outcome of patients who received CI.

Pt.no.	IAC malformation	Laterality of CI	Description	Post-OP CAP-II*
1	Type IIIb right, Type IV left	Unilateral/right ear	Not available for follow up	n/a
2	Type IIIb both sides	bilateral		8
3	Type IIIb right	Unilateral/right ear		8
4	Type IIIb right	Unilateral/right ear	Not available for follow up	n/a
5	Type IIIa both sides	bilateral		9
6	Type I right	Unilateral/right ear	Not available for follow up	n/a
7	Type I left	Unilateral/left ear		9
8	Type I left	Unilateral/left ear		7
9	Type I both sides	Unilateral/right ear		6
10	Type I both sides	Unilateral/left ear		6

*Categories of Auditory Performance II (CAP-II) Score.

In this context, it is also worth mentioning the increased incidence of syndromic disorders associated with impaired organogenesis. According to the results of the present study, over 25 per cent of patients examined also have a syndromic disorder. These include Goldenhar syndrome, Kippel-Feil syndrome, and Nager syndrome. In one case, trisomy 21 was also diagnosed. The scientific literature indicates that Branchio-Oto-Renal Spectrum Disorder (BORSD) is characterized in particular by malformations of the outer, middle, and inner ear. These malformations can be accompanied by conductive, sensory, or mixed hearing loss, fistulas and kidney malformations ranging from mild renal hypoplasia to bilateral renal agenesis.

In the present study, cochlear implantation was performed on ten patients (► **Table 1**). In two patients, an attempt at implantation was made, which led to facial nerve activation. In addition to cochlear implantation, postoperative hearing and speech rehabilitation programs are essential for developing hearing performance in all patients who receive a cochlear implant [24, 25, 26, 27]. Patients with both ABI and CI showed clear benefits from the implanted hearing system during the follow-up period. Patients who received a cochlear implant showed the greatest progress in terms of auditory performance (CAP score II; ► **Table 7**), with some achieving results comparable to individuals without hearing loss.

The classification models for IAC anomalies established by Jackler et al. and Senneroglu et al. each address types of IAC anomalies, but they do not consider them in isolation rather always in conjunction with other associated IAC anomalies. The present study overcomes the limitations of the current classification system and provides a more definitive classification of congenital anomalies of the IAC based on a large collective of 36 patients with isolated IAC malformation. Attributed to high-resolution MRI, evaluation of the vestibulocochlear nerve and the facial nerve was performed, similar to the patient series of Wang et al. [2].

In conclusion, due to the chameleon-like variability of IAC malformations, a strict and rigid classification is not useful. A basic classification into four groups as proposed in our study combined with careful assessment of the presence of the individual nerves may support clinical decision-making. For this purpose, we recommend the following diagnostic procedure 1 – evaluate the shape of the IAC (narrow-atresic-double with/without connection), 2 – separate assessment of the course and presence of the facial nerve, 3 – record the number (and then allocation) of nerve fibers in the cpa and the further course of the IAC.

Conclusion

We placed the malformations into four distinct groups, based on evidence of recurrent patterns. IAC malformations are often associated with vestibulocochlear and/or facial nerve malformations. Complete aplasia of the facial nerve is also possible and is clinically confirmed by facial nerve palsy. Malformations of the IAC are usually isolated from labyrinthine malformations, but may be part of syndromic diseases and may be accompanied by more profound malformations of the cranial nerve nuclei. Accurate analysis of the imaging data and knowledge of the complexity of the

interacting malformations is of great importance in assessing the likelihood and benefit of hearing system implantation, such as cochlear implant to treat related hearing loss.

Conflict of Interest

The authors declare that they have no conflict of interest.

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